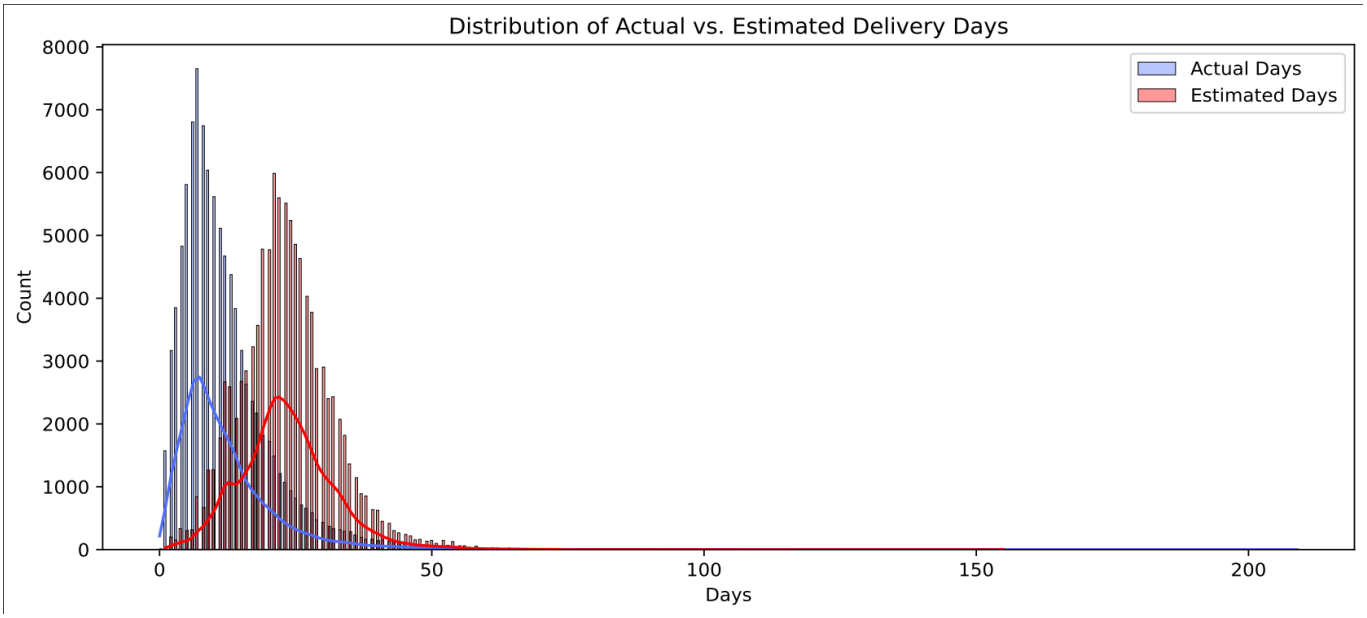
Delivery Performance SLA Analytics

What is SLA (Service Level Agreement)?

- A **formal agreement** between:
  - Service provider (e.g., company, seller, logistics partner)
  - Customer (or another business)
- Defines the **expected level of service**

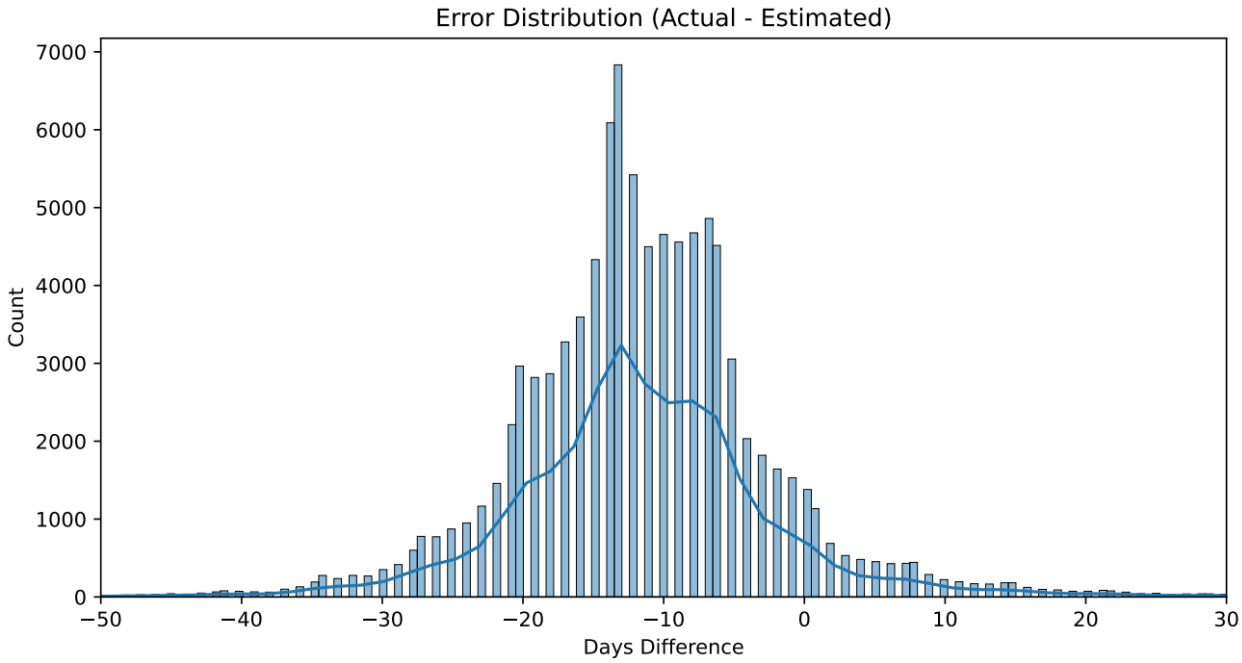
Early Orders	Late Orders	On Time Orders
87788	7308	1380
90.99%	7.57%	1.43%



Insights:

Conservative Estimation: Olist consistently provides a longer estimated delivery date than is typically required. While this may seem less competitive, it leads to ~85-91% of orders being delivered "early," which is strongly correlated with higher review scores.

The "Long Tail": Both distributions have long tails extending past 50 days. In the Olist dataset, these outliers often represent external factors like the major postal strikes that occurred in 2017 and 2018.

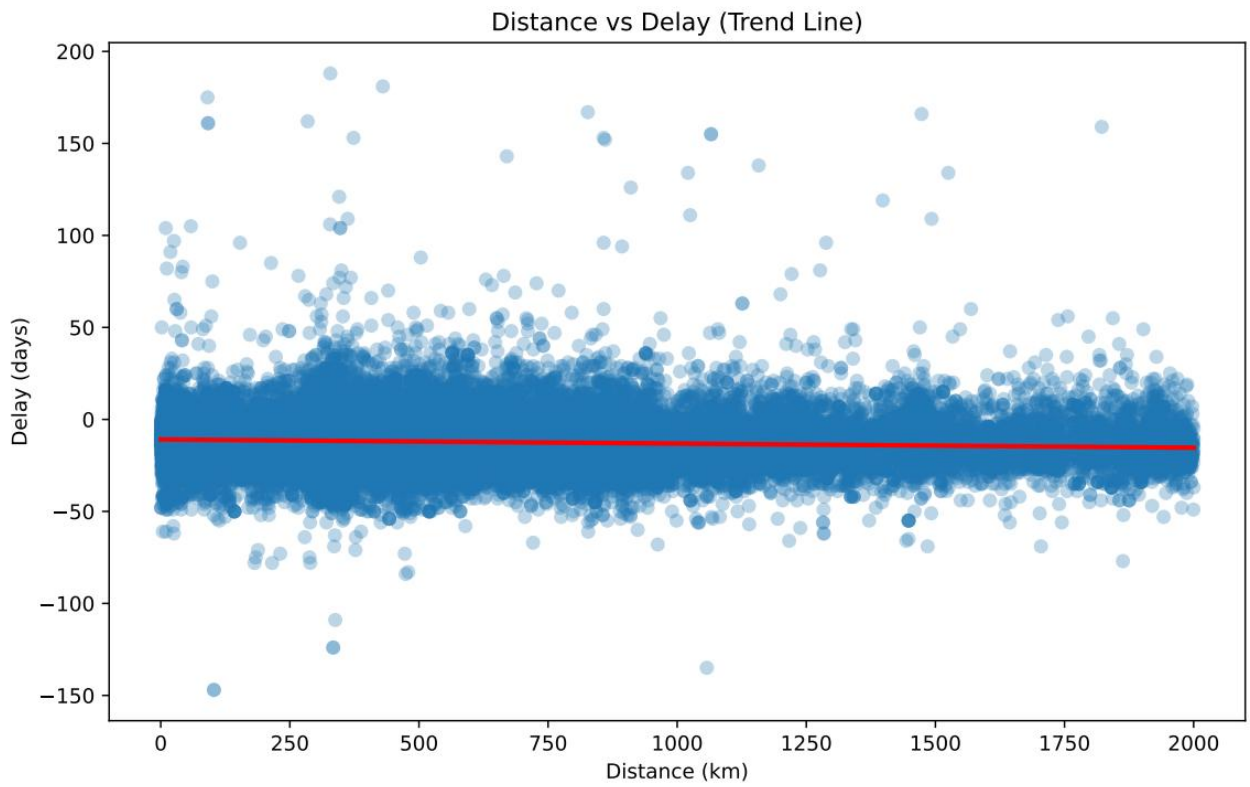


The Meaning of the Zero Line

- **0 (Center):** Deliveries that arrived exactly on the estimated day.
- **Negative Values (Left of 0):** These represent **early deliveries**. For example, a value of -10 means the package arrived 10 days earlier than promised.
- **Positive Values (Right of 0):** These represent **late deliveries**. These are the "late" orders that typically result in lower customer satisfaction scores.

Insights

- **Highly Conservative Estimates:** Notice that the vast majority of the data (the tall bars and peak) sits between **-20 and -5 days**. This confirms that Olist’s logistics model intentionally overestimates delivery times.
- **The "Safety Buffer":** By promising a delivery in, say, 25 days but delivering it in 12, Olist creates a "positive surprise" for the customer. In this dataset, roughly **85-90% of orders** are delivered before the estimated date.
- **The "Danger Zone":** The small tail to the right of 0 shows the orders that missed their deadline. Even though they are a minority, in the Olist dataset, these specific orders are the primary driver of **1-star reviews**.



1. The Trend Line (Red)

The most striking feature is that the red trend line is almost perfectly flat and slightly below the zero mark.

2. The Negative "Cloud" (The Bulk of Data)

Most of the blue dots are concentrated below the 0 line. This reinforces what we saw in the histograms: regardless of whether the customer is next door or across Brazil, they are likely to receive their package 10 to 15 days earlier than the official estimate.